

Aggregating Large-Scale Residential Users for Regulation Reserve Provision: Truthful Combinatorial Auction Based Approach

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Abstract—The proliferation of flexible loads has empowered residential users in contributing both upward and downward frequency regulation reserves to the power grid. Due to barriers like minimum bid size, residential users with relatively small power consumption must be aggregated before they can enter the reserve market. However, the large number of residential users renders the aggregation mechanism design problem a challenging task. In particular, the cost for reserve provision of each user is not only highly heterogeneous, but also coupled through both the temporal dimension and the up/down regulation direction. In order to allow the users to fully express their costs, in this paper, we propose a combinatorial reverse auction (CRA) framework as the market mechanism for user aggregation. In this auction, the aggregator is the auctioneer and procures reserves from residential users. The users submit package bids consisting of combinations of both upward and downward reserves over multiple time-slots, capturing the coupling in the user cost. Furthermore, to address the infamous computational challenges of large-scale combinatorial auctions, we develop a novel fast combinatorial auction (FCA) mechanism that can be solved in polynomial time. It includes an approximate winner determination algorithm and a critical payment scheme.

Notably, our proposed mechanism is rigorously proved to possess desirable economic properties such as truthfulness and individual rationality. Extensive simulations have validated the theoretic properties of the proposed CRA mechanism and its advantages over existing methods. In particular, compared with the widely employed truthful Vickrey-Clarke-Groves (VCG) mechanism, CRA can be 10^4 times faster than VCG when the user number is above 1000. Meanwhile, it is able to achieve near-optimal social cost, where the average optimality loss is 2.92%.

Index Terms—Regulation reserve, demand-side aggregation, combinatorial auction, mechanism design.

I. INTRODUCTION

A LONG with the high penetration of renewable energy sources, demand-side participation in ancillary services has been an active research field for the last decade [1]. For example, residential flexible loads such as heating ventilation and air conditioning (HVAC) devices and electric vehicles (EV) are able to provide frequency regulation reserves to the power system [2], [3]. However, due to barriers like minimum bid size and minimum time duration, residential users with relatively small power and energy consumption cannot enter the market directly, instead they need to be aggregated by an “aggregator” before joining the market [4].

Various previous works have investigated the aggregation problem of demand-side users to participate in the power markets. Depending on how the aggregator engages with demand-side users, two kinds of problems are studied. The first category assumes that the aggregator represents the users and thus has full information from users [5], [6], [7], [8], [9]. However, this assumption raises privacy concerns and undermines the users’ autonomy in decision-making. In contrast, another category posits that both the aggregator and users function as independent entities, with each making decisions autonomously [10], [11], [12], [13], [14], [15]. This scenario gives rise to a hierarchical optimization problem. Solving it requires accurate mathematical models of residential users, e.g., dynamics of flexible loads and user preference functions. In practice, this poses a significant challenge since user behavior can be too complex to be captured exactly with tractable mathematical formulations. The large-scale residential users are highly heterogeneous in terms of their load characteristics, physical constraints and preferences, further increasing the difficulty of accurate modeling.

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As a result, an auction that elicits information from heterogeneous residential users in the form of bids [16] is a more appropriate solution. In this way, users are relieved of providing exact mathematical models to the aggregator. Instead, users can simply submit a bid to the aggregator, indicating the reserve they can provide and the corresponding cost. Hence, the auction framework is flexible enough to accommodate heterogeneous users with mixed rationality behaviors, and also protects user privacy. Meanwhile, the aggregator can determine the procurement outcome and payments based solely on the received bids, without the need for solving complex bilevel optimization problems. Furthermore, the autonomy of users is preserved in auctions as they can voluntarily decide whether or not to participate in the reserve market.

Despite the advantages of implementing auction as the market mechanism, we will confront three major obstacles when designing auctions for residential user aggregation in regulation reserve provision. First, reserve provision through flexible loads incurs cost such as discomfort or inconvenience to the user. As we will further discuss in Section III-D, due to the coupling between upward regulation reserve (up-reserve) and downward reserve (down-reserve), together with the coupling of reserves over the temporal dimension, submitting the combination of up and down reserves over several time-slots as a package can better express the preference of users. Hence, a combinatorial auction (CA) which allows package bids should be introduced. Second, users' costs are private and unknown to the aggregator, then a user may perform strategically and report false value to manipulate the market for her own good. Thus, designing a *truthful* auction, where a user can achieve maximum utility by simply bidding her true cost regardless of what other users act, is crucial in ensuring the market efficiency, fairness and robustness. In addition, to attract more users, this auction should also be *individually rational*, where the user utility of participating in the auction must be non-negative. Third, considering the NP-hardness of CA's winner determination problem and the large number of residential users to be aggregated,¹ the implementation of the auction mechanism faces significant computation requirements.

To this end, we introduce a novel combinatorial reverse auction (CRA) mechanism,² where an aggregator procures upward and downward regulation reserve from large-scale residential users based on their package bids. This auction is tailored for tackling the three obstacles discussed above, with the main contributions summarized as follows:

- We introduce a CRA for the user aggregation problem. This auction format allows package bidding to fully capture residential users' preferences in providing regulation reserve.
- Due to the NP-hardness of the winner determination problem (WDP) in combinatorial auctions, the computational complexity increases exponentially with the number of bidders. In order to solve this scalability challenge brought

by large number of residential users, we propose a near-optimal approximation algorithm to solve the WDP, as well as a critical payment scheme to compute the payments for bidders. Together they constitute a computationally efficient CRA mechanism by iteratively seeking the cost-effective bids, instead of solving the NP-hard integer programming problem directly.

- Based on rigorous theoretical analyses, we prove that the proposed mechanism possesses crucial economic properties such as truthfulness and individual rationality. We also prove that the mechanism can be implemented in polynomial time, verifying its computational efficiency from theory.

The rest of this paper is structured as follows. Section II discusses the related literature. Section III describes the user model and Section IV presents the desired framework of reserve auction. Then, a CRA mechanism is proposed in Section V, followed by theoretical analysis of its properties. Numerical results are presented in Sections VI, and VII concludes the paper.

II. RELATED WORKS

There have been an increasing amount of works to apply auction theory in constructing emerging power markets, where demand-side users are encouraged to trade energy and flexibility for ancillary services. For example, in [17] and [18], the procurement auctions for demand response (DR) service are studied in an online reverse auction mechanism and a randomized dynamic programming based auction mechanism, respectively. In [19], Vickery-Clarke-Groves (VCG)-based incentive mechanisms are introduced for the emergency DR procurement in multi-tenant buildings. In [20], Ausubel's clinching auction is applied for the DR problem, in order to achieve a distributed implementation of the VCG mechanism. Enabled by the blockchain technology, [21] consolidates bilateral contracts and double auction mechanism into a local energy market. Authors of [22] leverage an incentive compatible mechanism based on VCG auction in a reactive power ancillary service market to pursue economic efficiency. In [23], a deep learning based method is applied to automatically improve VCG for tackling the budget balanced local energy market. In [24], a second-price sealed-bid auction is proposed to coordinate EV charging time shift to reduce the photovoltaic power curtailment. In [25], a continuous group-wise double auction mechanism is developed to coordinate the energy transactions among prosumers in distribution-level markets. In [26], energy transactions of agents in distribution-level markets are performed according to second-price sealed-bid auctions.

Different from the auction formats reviewed above, the user bids for regulation reserve need to be modeled as package bids. This leads to a combinatorial auction (CA), which requires different methods of mechanism design.

In fact, CA has already drawn attentions from the power system community. In [27], vehicle-to-grid CAs are proposed which use EV charging/discharging powers to regulate voltage deviations of distribution system. To elicit truthful bids from generators, an incentive mechanism is developed in [28] to clear the energy and reserve market in multi-area power systems.

¹The power consumption of residential user is about several kilowatts, while the minimum bid size of reserve market is in general several megawatts.

²The reverse auction is an auction where the bidders are sellers. If the bidders are buyers, it is a forward auction.

Both [27] and [28] select the widely used VCG mechanism. Although VCG mechanism is versatile and truthful, it is not suitable for large-scale problems. As discussed in later sections, VCG mechanism needs to solve an NP-hard WDP to exact optimum, which violates the computation requirement when a large number of users participate in the auction.

To relieve the computation burden of exactly solving the WDP, researchers have explored various approximate algorithms for CA. In both [29] and [30], metaheuristic methods such as genetic algorithm and particle swarm optimization are used to study the CA in energy storage sharing and microgrids energy trading, respectively. The authors of [31] designed a linear programming (LP)-based CA mechanism with expressive bids for prosumers in a local energy market. In [32], a truthful fully polynomial time approximation scheme is proposed to solve the NP-hard WDP in the CA for energy storage sharing. Despite the social welfare is optimized, the constraints of WDP may be violated. Ahmed et al. [33] utilize graph neural networks (GNN) to construct a new CA framework for local energy flexibility markets. However, similar with [32], [33] cannot ensure the satisfaction of WDP constraints.

Another stream of research for CA is the iterative mechanisms. In [34], a staggered clock-proxy auction mechanism for the DR problem is developed. In [35], an iterative CA mechanism is introduced for the energy trading problem of flexible prosumers. However, optimization problems need to be solved by each bidder in each iteration, limiting their scalability and computation tractability when the bidder valuation functions are non-convex and the number of participating bidders is large.

To summarize, different from existing works on CA application in the power system, this work proposes a CA mechanism in the reverse auction setting, which satisfies WDP constraints and achieves near-optimal social cost with high computation efficiency, and possesses desirable economic properties such as truthfulness and individual rationality.

III. RESIDENTIAL USER MODEL

We consider the problem of aggregating large numbers of residential users to provide frequency regulation reserves in real-time, including both up-reserve and down-reserve. Three kinds of representative flexible residential loads are introduced, including deferrable loads (DL), heating, ventilation, and air conditioning (HVAC) devices and electric vehicles (EV). $\mathcal{T} = \{1, 2, \dots, T\}$ is the time horizon for consideration.

A. Deferrable Loads

Firstly, we consider deferrable loads such as washing machines, where the discomfort cost of reserve provision depends on its total load adjustment ΔD [36]:

$$\Delta D = \sum_{t=1}^T (q_t^d - q_t^u), \quad (1)$$

where q_t^u and q_t^d are the up and down reserves respectively.

The discomfort cost is

$$C_{DL}^{dis} = \begin{cases} h_{dl} |\Delta D| + \sum_t c_{dl}^0 \mathbf{1}(\max(q_t^d, q_t^u)), & \text{if } \Delta D \leq 0 \\ \sum_t c_{dl}^0 \mathbf{1}(\max(q_t^d, q_t^u)), & \text{otherwise.} \end{cases} \quad (2)$$

When $\Delta D > 0$, the cost mainly comes from the inconvenience incurred by not freely using her own device due to reserve provision, where $c_{dl}^0 > 0$ is the inconvenience cost coefficient. When $\Delta D \leq 0$, aside from the inconvenience cost, there is also cost from the discomfort of not finishing the task, $h_{dl} \geq 0$ is the discomfort cost coefficient. The function $\mathbf{1}(x)$ is the indicator function, which only equals to one when either up-reserve or down-reserve is non-zero. Hence, the total cost is

$$C_{DL} = C_{DL}^{dis} + \sum_t h_{ep}(t)(-q_t^u + q_t^d), \quad (3)$$

where $h_{ep}(t)$ is the electricity energy price.

B. HVAC

Secondly, we focus on the HVAC device. For a residential user possessing an HVAC, the indoor temperature at time-slot t is denoted as T_t , while the ambient temperature outside is set as T_t^{amb} . The power demand of the HVAC is d_t . The thermal dynamics of the household is as follows [37]:

$$T_{t+1} = (1 - \alpha)T_t + \alpha T_t^{\text{amb}} - \beta d_t, \quad (4)$$

where α and β are constant coefficients related to thermal capacity and thermal resistance.

Assume that the indoor temperature is fixed at T_{ref} when the user is not providing any reserve. The temperature deviation at each time-slot t due to reserve provision q_t^u and q_t^d is

$$\Delta T_t = \beta \sum_{\tau=0}^{t-1} (1 - \alpha)^\tau (q_{t-\tau}^u - q_{t-\tau}^d), t \geq 1. \quad (5)$$

The quadratic function has been widely used in previous literature [38] to model the user's discomfort due to deviation from desired temperature T_{ref} . Therefore, the temperature deviation from providing regulation reserves causes discomfort to the residential user:

$$C_{HVAC}^{dis} = \sum_t h_{hvac} \Delta T_t^2. \quad (6)$$

where h_{hvac} is the discomfort coefficient. Then, the total cost becomes

$$C_{HVAC} = C_{HVAC}^{dis} + \sum_t c_{hvac}^0 \mathbf{1}(\max(q_t^d, q_t^u)) + \sum_t h_{ep}(t)(-q_t^u + q_t^d), \quad (7)$$

where the second term models the inconvenience whenever reserve is provided, $c_{hvac}^0 > 0$ is the inconvenience cost coefficient of HVAC. Similar with the case in DL, the last term is the energy cost due to reserve activation.

C. Electric Vehicle

Thirdly, electric vehicle (EV) batteries are studied. The battery state of charge (SOC) at time-slot t is

$$E_t = E_0 + \sum_{\tau=1}^t \frac{1}{B_{\max}} \left(\eta^c q_{\tau}^d - \frac{q_{\tau}^u}{\eta_d} \right), \quad (8)$$

where E_0 is the initial SOC, $B_{\max}$ is the battery capacity, η_c and η_d are charging and discharging efficiencies, respectively. Using the same reasoning with reference [39], the battery degradation cost due to reserve provision is defined as follows:

$$C_{dg} = \sum_{t=0}^{T-1} C_{d,t}(E_t, E_{t+1}), \quad (9)$$

where the function $C_{d,t}$ is

$$C_{d,t} = \begin{cases} \frac{C_{inv}}{N_{cycle}(E_{t+1})} - \frac{C_{inv}}{N_{cycle}(E_t)}, & E_t > E_{t+1} \\ 0, & \text{otherwise} \end{cases} \quad (10)$$

here N_{cycle} is a function that represents the lifetime cycle numbers of batteries. Its relationship with the battery depth-of-discharge (DoD) can be modeled by the following expression:

$$N_{cycle}(DoD) = \gamma_0 \times DoD^{-\gamma_1} \times e^{\gamma_2(1-DoD)}, \quad (11)$$

and the DoD satisfies $DoD = 1 - SoC$. Constants γ_0 , γ_1 and γ_2 are constant parameters.

Note that deviation from desired SOC may bring discomfort to EV owners, since the owner may need to travel when the time horizon ends. Suppose the original power consumption plan can achieve the desired SOC, then, similar to [40], this discomfort cost is defined as follows:

$$C_{EV}^{dis} = h_{ev}(E_T - E_{desire})^2 \quad (12)$$

$$= \frac{h_{ev}}{B_{\max}^2} \left(\sum_{t=1}^T \left(\eta^c q_t^d - \frac{q_t^u}{\eta_d} \right) \right)^2, \quad (13)$$

where h_{ev} is the discomfort coefficient.

Now the cost of electric vehicles for providing reserves is defined as

$$C_{EV} = C_{dg} + C_{EV}^{dis} + \sum_t h_{ep}(t)(-q_t^u + q_t^d) + \sum_t c_{ev}^0 \mathbf{1}(\max(q_t^d, q_t^u)), \quad (14)$$

where the third term is the cost incurred by the adjustment of the electrical energy. The last term models the inconvenience of not freely using EV, and c_{ev}^0 is the inconvenience cost coefficient.

D. Motivation for Package Bidding

Based on the models of aforementioned flexible loads, residential users submit reserve bids to the aggregator to participate in the auction. Each user's bid includes the reserve they can provide and the incurred cost. Details of the auction will be discussed in the next section. After the auction is complete, the winning user will receive a payment to cover their incurred cost. In this subsection, we present simple motivating examples to

showcase the importance of package bids for users. A package bid is a reserve bid that includes both up and down reserves over the time horizon.

Here we only study two time-slots for a user with HVAC. If the up-reserve q_1^u and q_2^u at both time-slots are called upon, according to (4), the temperature deviations in both time-slots are

$$\Delta T_1 = \beta q_1^u, \quad (15)$$

$$\Delta T_2 = (1 - \alpha)\beta q_1^u + \beta q_2^u, \quad (16)$$

the incurred discomfort cost is

$$h_{hvac}[(1 + (1 - \alpha)^2)\beta^2(q_1^u)^2 + \beta^2(q_2^u)^2 + 2(1 - \alpha)\beta^2 q_1^u q_2^u]. \quad (17)$$

Equation (17) means that the discomfort cost due to temperature deviation is temporally coupled through the term consisting of both q_1^u and q_2^u . Therefore, when a user is making a bidding decision about the amount of reserve to provide, she should consider its effect on future decisions. In other words, combining the reserve bids across several time-slots as a package can better reflect the true discomfort cost of a user.

On the other hand, with the trend of relaxing constraints on symmetric up-reserve and down-reserve [41], the decision of providing either up or down reserve can be performed independently. However, the discomfort costs incurred by up and down reserve provision are still coupled. For example, for a user with an EV, when she is committed to provide up-reserve, she needs to keep her EV plugged and cannot drive it freely. Then, no matter down-reserve is provided or not, the inconvenience caused by the EV plugging constraint is still the same. Hence, a bid package composed of both up and down reserve can capture the user inconvenience more accurately than bidding them separately.

To summarize, in order to fully express the reserve cost of residential users, the temporal coupling together with the coupling between up and down reserve lead to a need for package bidding.

IV. RESERVE AGGREGATION AUCTION

This section first presents the basic framework of the combinatorial reverse auction to aggregate reserves from large-scale users, and then introduces the problem formulation and design requirements for this auction.

A. CRA Framework

The aggregator procures up and down regulation reserve through aggregating small reserve capacities from a set of residential users $\mathcal{K} = \{1, 2, \dots, K\}$.

Each user $k \in \mathcal{K}$ submits a package bid $\beta_k = (Q_k, b_k)$ to the aggregator for reserve provision. $Q_k = (q_{1,k}^u, q_{1,k}^d, \dots, q_{t,k}^u, q_{t,k}^d, \dots, q_{T,k}^u, q_{T,k}^d)$ is the provided up-reserve and down-reserve over the time horizon $\mathcal{T} = \{1, 2, \dots, T\}$, where $q_{t,k}^u$ and $q_{t,k}^d$ are the up-reserve and down-reserve at time-slot t , respectively. b_k is the reported cost of user k for providing reserve.

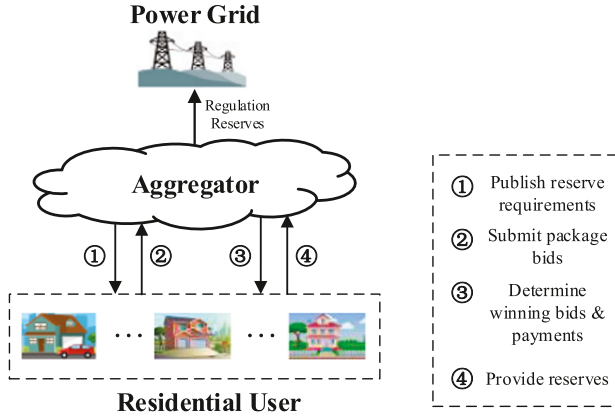


Fig. 1. Auction framework for aggregation of regulation reserves.

Note that a user k can act strategically and report a fake cost b_k that is different from the true cost $\bar{b}_k$, where $\bar{b}_k$ is equal to C_{DL} , C_{HVAC} or C_{EV} , depending on which kind of device this user possesses. The definitions of C_{DL} , C_{HVAC} and C_{EV} can be found in equation (3), (7) and (14), respectively. Later we will elaborate on how to design an auction mechanism that encourages users to report true cost and thus avoid their strategic behaviors.

The reserve requirements of the aggregator constitute a set $\mathcal{L} = \{L_1^u, L_1^d, L_2^u, L_2^d, \dots, L_T^u, L_T^d\}$, where L_t^u and L_t^d are the up-reserve and down-reserve requirements at time-slot t , respectively. After receiving the package bids from residential users, the aggregator performs a CRA to determine which user will be selected for reserve provision and how much payments should be provided to the winning users. As presented in Fig. 1, the basic framework of the CRA mechanism is implemented as follows:

- 1) The aggregator publishes reserve requirement information to the residential users.
- 2) Each user $k \in \mathcal{K}$ submits her own bid to the aggregator, which is $\beta_k = (Q_k, b_k)$, Q_k represents the provided up-reserve and down-reserve over the time horizon $\mathcal{T}$, and b_k is her reported cost for providing reserve.
- 3) After receiving the bids, the aggregator determines which bids win the auction, as well as the payments to the winning users.
- 4) Winning users receive payments from the aggregator by providing reserves.

Remark Here we adopt a common assumption in auction theory that the allocation dictated by the auctioneer is final, where no post-auction actions from users are allowed [42]. Otherwise, the violating user will be punished, e.g., paying a fine or denying access to future auctions.

Within the CRA framework, determining the winning bids and calculating the corresponding payments are the most important tasks in designing the auction, which will be our focus from now on.

B. Problem Formulation

The mathematical formulations of CRA mechanism design include two parts: winner determination and payment

calculation, where each of them will be specified in this subsection.

At first, the aggregator needs to determine the auction winners, which is formulated as the following winner determination problem (WDP):

$$\text{WDP: } \min \sum_{k \in \mathcal{K}} b_k x_k \quad (18a)$$

$$\text{s.t. } \sum_{k \in \mathcal{K}} q_{t,k}^u x_k \geq L_t^u, \forall t \in \mathcal{T} \quad (18b)$$

$$\sum_{k \in \mathcal{K}} q_{t,k}^d x_k \geq L_t^d, \forall t \in \mathcal{T} \quad (18c)$$

$$x_k \in \{0, 1\} \quad (18d)$$

where the binary variable $x_k = 1$ represents that user k is the winner of this auction, while $x_k = 0$ means k is the loser. The objective function (18a) minimizes the social cost of reserve procurement, constraints (18b) and (18c) enforce that reserve requirements are satisfied by reserves aggregated from users. However, as shown in Theorem 1, solving this integer program WDP is not an easy task:

Theorem 1: WDP is an NP-hard problem.

Proof: Set all the $b_k, q_{t,k}^u, q_{t,k}^d, L_t^u$ and L_t^d in (18) to 1, problem (18) reduces to an NP-hard set cover problem [43]. Therefore, as a generalization of the set cover problem, WDP is also NP-hard. ■

Remark b_k in (18a) is the cost claimed by the user herself, which may not be the true cost. In order to promote market efficiency and fairness by truthful bidding, the following sections will pursue an auction mechanism that can guarantee submitting true cost to the aggregator is an optimal solution to the users. Thus, b_k is indeed the true cost $\bar{b}_k$.

As for the payments to users, we seek to design payment schemes so that the resulting auction satisfies desirable properties, e.g., truthful and individual rational. Before introducing the formal concept of those properties, we bring in the definitions of user utility and dominant strategy first:

Definition 1: User k 's utility u_k is equal to her received payment p_k subtracted by her true cost, i.e.,

$$u_k = p_k - \bar{b}_k. \quad (19)$$

Definition 2: Bidding β_k^* is defined as the dominant strategy of user k , if for any alternative strategy β_k of user k and any strategy profile of other users β_{-k} , we have:

$$u_k(\beta_k^*, \beta_{-k}) \geq u_k(\beta_k, \beta_{-k}). \quad (20)$$

Then we introduce the definitions of truthfulness and individual rationality:

Definition 3: An auction mechanism is truthful if for every user $k \in \mathcal{K}$, it is her dominant strategy to bid the true cost $\bar{\beta}_k = (Q_k, \bar{b}_k)$. Namely,

$$u_k(\bar{\beta}_k, \beta_{-k}) \geq u_k(\beta_k, \beta_{-k}), \forall \beta_k, \forall \beta_{-k}. \quad (21)$$

Definition 4: An auction mechanism is individually rational if every participating user has non-negative utility, i.e., $u_k \geq 0, \forall k \in \mathcal{K}$.

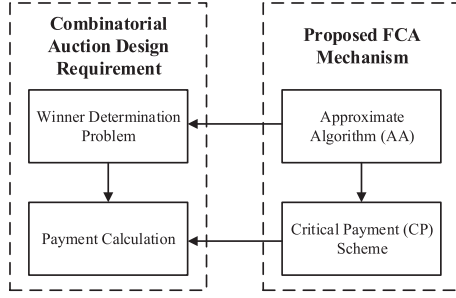


Fig. 2. Two components of the proposed FCA mechanism. An approximate algorithm (AA) is first proposed to solve the WDP of CA, and then a critical payment (CP) scheme is developed to calculate the payments for winning bidders.

According to the definitions, satisfying truthfulness means that no user has incentive to report a fake cost value, as reporting the truth already achieves the optimal utility. Meanwhile, if an auction satisfies individual rationality, then every user who participates in it cannot be worse than not participating.

Remark A conventional way to design a truthful combinatorial auction is to apply the VCG mechanism. The VCG mechanism is also “the only well understood general method for the design of truthful mechanisms” [44]. However, a prerequisite for implementing VCG is to exactly solve the WDP. If we can only obtain an approximate solution of WDP, using VCG payment scheme cannot yield a truthful mechanism [45]. As shown in Theorem 1, there does not exist a polynomial-time solvable algorithm to obtain the exact global optimum of the large-scale WDP (18). Consequently, VCG is not an appropriate mechanism for the large-scale user aggregation scenario. Hence, we have to seek a non-VCG mechanism that satisfies truthfulness, individual rationality, and can be solved efficiently.

V. AUCTION MECHANISM DESIGN

In this section, we first devise an approximate algorithm for the WDP, and then propose a novel payment calculation scheme. Together they constitute a computationally efficient auction mechanism, named *fast combinatorial auction* (FCA), as shown in Fig. 2. Lastly, we prove that FCA has desirable properties such as truthfulness and individual rationality via rigorous theoretical analyses.

A. Approximate Algorithm for WDP

Since WDP is an NP-hard problem, the first step to design FCA is to develop an approximate algorithm (AA) instead of solving it exactly.

The intuition of AA is to iteratively choose the most “cost-effective” bid in a greedy manner. Here, “cost-effectiveness” of each bid k is defined by the rank value $r(\beta_k)$ in (22). In (22), $\tilde{L}_\tau^u$ and $\tilde{L}_\tau^d$ are the unserved up-reserve and down-reserve demands for time-slot τ in the current iteration, respectively. The term $\min\{\tilde{L}_\tau^u, q_{\tau,k}^u\}$ and $\min\{\tilde{L}_\tau^d, q_{\tau,k}^d\}$ represent the contribution of bid k in serving the unsatisfied up-reserve and down-reserve of time-slot τ in the current iteration, respectively. If $\tilde{L}_\tau^u \geq q_{\tau,k}^u$, this term equals to $q_{\tau,k}^u$, i.e., the reserve bid of k can be fully accounted as the contribution to serve reserve

Algorithm 1: Approximate Algorithm (AA) for WDP.

Input: The set of up, down reserve demands $\mathcal{L}^u$ and $\mathcal{L}^d$; the bids set $\mathcal{B}$;

Output: Winner set $\mathcal{W}$;

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1:  $\mathcal{W} \leftarrow \emptyset, \tilde{L}_t^u \leftarrow L_t^u, \tilde{L}_t^d \leftarrow L_t^d$ ;
2: while  $\sum_{t \in \mathcal{T}} \tilde{L}_t^u \neq 0 \vee \sum_{t \in \mathcal{T}} \tilde{L}_t^d \neq 0$ 
3:   for all  $\beta_j$  in  $\mathcal{B}$ 
4:     Calculate the rank value  $r(\beta_j)$  of bid  $\beta_j$  according to (22).
5:   end for
6:   Sort  $r(\beta_j), \forall \beta_j \in \mathcal{B}$  nondecreasingly, and use  $\mathcal{R}$  to denote the sorted list;
7:   Let the head of list  $\mathcal{R}$  be  $\beta_h = (Q_h, b_h)$ ;
8:    $\mathcal{W} \leftarrow \mathcal{W} \cup \beta_h$ ;
9:   for all  $t$  s.t.  $(q_{t,h}^u, q_{t,h}^d) \in Q_h$  do
10:     $\tilde{L}_t^u \leftarrow \tilde{L}_t^u - \min\{\tilde{L}_t^u, q_{t,h}^u\}$ ;
11:     $\tilde{L}_t^d \leftarrow \tilde{L}_t^d - \min\{\tilde{L}_t^d, q_{t,h}^d\}$ ;
12:   end for
13:   Delete  $\beta_h$  from  $\mathcal{B}$ ;
14: end while
15: return  $\mathcal{W}$ ;
```

demand. If $\tilde{L}_\tau^u < q_{\tau,k}^u$, this term equals to $\tilde{L}_\tau^u$, i.e., the proportion of reserve capability exceeding the current unserved reserve demand will not be considered as bid k ’s contribution. Therefore, the denominator of (22) means the total contribution of bid k to the unserved up-reserve and down-reserve, while the numerator of (22) is the cost of bid k . Then the rank value $r(\beta_k)$ represents the average cost of bid k for each unit of reserve contribution. In other words, it is the “price” of user k ’s bid. Note that the “price” is not an intrinsic value of a bid, since it may be related to the current unsatisfied reserve demands.

$$r(\beta_k) = \frac{b_k}{\sum_{\tau: (q_{\tau,k}^u, q_{\tau,k}^d) \in Q_k} (\min\{\tilde{L}_\tau^u, q_{\tau,k}^u\} + \min\{\tilde{L}_\tau^d, q_{\tau,k}^d\})}. \quad (22)$$

Details of AA are elaborated in Algorithm 1, where the winning bids are determined iteratively. In (Step 1) of Algorithm 1, the set of winning bids $\mathcal{W}$ and the unserved reserve demands are initialized. Algorithm 1 iterates until the unserved reserve demands become zero (Step 2). In each iteration, the “cost-effectiveness” of each remaining bid is calculated (Step 4) and sorted into a list $\mathcal{R}$ (Step 6). The head of $\mathcal{R}$ (Step 7) β_h is added to the winner set, since it is the most “cost-effective” bid in the current iteration (Step 8). Finally, the unserved reserve demands in each time-slot t are served by β_h , and thus are updated based on β_h ’s contribution (Step 10 & 11). After that, β_h is also removed from the set of remaining bids (Step 13).

B. Payment Scheme Design

After Algorithm 1 has determined the winners, we propose a payment scheme based on the idea of *critical payment* [46]. It is extended to the reverse auction setting where the WDP can be regarded as a generalization of the set-cover problem. In

Algorithm 2: Critical Payment (CP) Scheme for Winning Users.

Input: User k 's bid β_k , bids from other users

$\mathcal{B}_{-k} = \{\beta_j | j \neq k, j \in \mathcal{K}\};$

Output: Critical payment p_k^c , critical bid β_k^c ;

// Initialization

```

1:  $p_k^c \leftarrow 0, \tilde{L}_{t,k}^u \leftarrow L_t^u, \tilde{L}_{t,k}^d \leftarrow L_t^d;$ 
2: while  $\sum_{t \in \mathcal{T}} \tilde{L}_{t,k}^u \neq 0 \vee \sum_{t \in \mathcal{T}} \tilde{L}_{t,k}^d \neq 0$  do
3:   for all  $\beta_j$  in  $\mathcal{B}_{-k}$ 
4:     Calculate the rank value  $r(\beta_j)$  of bid  $\beta_j$  according
       to (22);
5:   end for
6:   Sort  $r(\beta_j), \forall \beta_j \in \mathcal{B}_{-k}$  nondecreasingly, and use  $\mathcal{R}^k$ 
       to denote the sorted list;
7:   Let the head of list  $\mathcal{R}^k$  be  $\beta_k^h = (Q_k^h, b_k^h);$ 
8:    $\mathcal{W}_k \leftarrow \mathcal{W}_k \cup \beta_k^h;$ 
9:   Calculate the critical payment candidate  $p_k^h$ 
       according to (23);
10:  if  $p_k^h > p_k^c$  then
11:     $p_k^c = p_k^h, \beta_k^c = \beta_k^h;$ 
12:  end if
13:  for all  $t$  s.t.  $(q_{t,h}^u, q_{t,h}^d) \in Q_k^h$  do
14:     $\tilde{L}_{t,k}^u \leftarrow \tilde{L}_{t,k}^u - \min\{\tilde{L}_{t,k}^u, q_{t,h}^u\};$ 
15:     $\tilde{L}_{t,k}^d \leftarrow \tilde{L}_{t,k}^d - \min\{\tilde{L}_{t,k}^d, q_{t,h}^d\};$ 
16:  end for
17:  Remove  $\beta_k^h$  from  $\mathcal{B}_{-k};$ 
18: end while
19: return  $\beta_k^c$  &  $p_k^c.$ 

```

contrast, the one in [46] is a forward auction whose WDP is a knapsack problem.

Details of the proposed payment scheme are shown in Algorithm 2. For each winner k , (Step 1) of Algorithm 2 first sets the critical payment p_k^c as zero, and then initializes the unserved reserve demands $\tilde{L}_{t,k}^u$ and $\tilde{L}_{t,k}^d$. Similar with Algorithm 1, Algorithm 2 iterates until the unserved reserve demands become zero (Step 2), and the “cost-effectiveness” of each remaining bid is calculated (Step 4) and sorted (Step 6). The difference is that here the iterations are over the set of all the bids excluding winner k . The remaining winner set after excluding k is denoted as $\mathcal{W}_k$ (Step 8). The “price” of each member of $\mathcal{W}_k$ is used to calculate the critical payment candidate p_k^h according to (23) (Step 9). The critical payment p_k^c is the largest value among all the candidate critical payments p_k^h . It is also the maximum cost value user k needs for winning the auction. Similar with Algorithm 1, (Step 13-17) are used to update the unserved reserve demands.

$$\begin{aligned}
p_k^h &= r(\beta_k^h) \cdot \sum_{\tau: (q_{\tau,k}^u, q_{\tau,k}^d) \in Q_k^h} (\min\{\tilde{L}_{\tau,k}^u, q_{\tau,k}^u\} + \min\{\tilde{L}_{\tau,k}^d, q_{\tau,k}^d\}) \\
&= b_k^h \frac{\sum_{\tau: (q_{\tau,k}^u, q_{\tau,k}^d) \in Q_k^h} (\min\{\tilde{L}_{\tau,k}^u, q_{\tau,k}^u\} + \min\{\tilde{L}_{\tau,k}^d, q_{\tau,k}^d\})}{\sum_{\tau: (q_{\tau,h}^u, q_{\tau,h}^d) \in Q_k^h} (\min\{\tilde{L}_{\tau,k}^u, q_{\tau,h}^u\} + \min\{\tilde{L}_{\tau,k}^d, q_{\tau,h}^d\})} \quad (23)
\end{aligned}$$

As for losing bids, the payments to them are zero. Therefore, the payments to all the users are summarized as follows:

$$p_k = \begin{cases} p_k^c, & \beta_k \in \mathcal{W} \\ 0, & \beta_k \notin \mathcal{W}. \end{cases} \quad (24)$$

C. Theoretical Analyses

The proposed AA algorithm for WDP and the critical payment scheme comprise the FCA mechanism. Here we provide rigorous proofs that FCA satisfies certain desirable properties. Before that, Algorithm 1 is proved to be monotone to facilitate the subsequent theoretic analyses. The definition of *monotonicity* is introduced as follows:

Definition 5: A mechanism is defined as monotone if for every bid $\beta_k = (Q_k, b_k)$, when β_k wins, a bid with lower bidding price $\hat{\beta}_k = (Q_k, \hat{b}_k), \hat{b}_k \leq b_k$ also wins.

Lemma 1: Algorithm 1 is monotone.

Proof: The proof is deferred to Appendix A. ■

Then, the notion of *critical value* is introduced in Lemma 2:

Lemma 2: For each winning user k , a **critical value** $c(\beta_k)$ exists, where user k wins if the cost value b_k she bids satisfies $b_k \leq c(\beta_k)$, and otherwise she loses.

Proof: The proof is deferred to Appendix B. ■

The critical payment to user k in FCA is exactly its critical value, as shown in Lemma 3:

Lemma 3: For winning user k , critical payment p_k^c and the critical value are the same.

Proof: The proof is deferred to Appendix C. ■

Lemma 4: The utility of a user who reports the true cost value is always non-negative when she received the critical payment.

Proof: Based on Lemma 2 and 3, when a truthful user k wins the auction, her cost value must satisfy that $\bar{b}_k \leq p_k^c$. Hence, her utility becomes $u_k = p_k^c - \bar{b}_k \geq 0$. As for the losing bidder, the utility is zero because both the received payment and her cost are zero. ■

Based on Lemma 2, 3 and 4, the following theorem follows:

Theorem 2: The FCA mechanism that consists of the proposed AA algorithm for WDP and the CP payment scheme is truthful.

Proof: The proof is deferred to Appendix D. ■

Truthfulness of FCA means that no user can increase her utility by submitting false bids, thereby discouraging the strategic behavior of dishonest bidding. This is crucial to ensure the market efficiency, fairness and robustness.

Furthermore, we can also prove the individual rationality of FCA:

Corollary 1: The FCA mechanism is individually rational.

Proof: This result directly follows from Definition 4, Lemma 4 and Theorem 2. ■

Individual rationality guarantees that each bidder can obtain non-negative utility, meaning that participating in the auction always results in an equal or improved utility compared to not participating. This encourages the participation of more users, as they are assured of a favorable outcome by joining the auction.

Finally, computational efficiency of the proposed FCA mechanism is shown by Theorem 3:

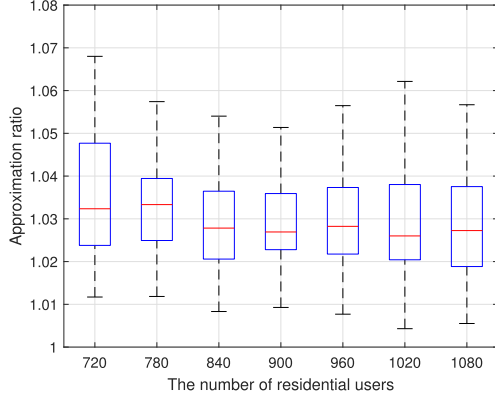


Fig. 3. Box plot of the approximation ratio of AA. The approximation ratio is the objective value of WDP achieved by AA divided by the optimal value.

Theorem 3: The FCA has polynomial-time computation complexity.

Proof: The proof is deferred to Appendix E. ■

VI. NUMERICAL RESULTS

Based on the models discussed in Section III, we consider residential users that each of them possesses one flexible load, i.e., deferrable load, HVAC or EV. The number of users with each kind of device is identical. The whole time horizon consists of $T = 4$ time-slots where each time-slot is 15 mins. The electricity price is $h_{ep} = 0.16$ \$/kWh. For users with deferrable loads, suppose their power consumption is uniformly distributed in [1,3] kW. The discomfort cost coefficient h_{dl} is uniformly distributed in [1.8,2.2] \$/kWh, and the inconvenience coefficient c_{dl}^0 is uniformly distributed in [0.025,0.05] \$. For users with HVAC, power consumption is uniform in [0,4] kW. Coefficients h_{hvac} and c_{hvac}^0 are uniform in [18,22] and [0.025,0.05] respectively. Thermal coefficients are set as $\alpha = 0.5013$ and $\beta = 0.1$ based on [38]. For users with EV, power consumption is uniform in [0,5] kW. Coefficients h_{ev} and c_{ev}^0 are uniform in [18,22] and [0.025,0.05] respectively. Charging/discharging efficiency η_c and η_d are both assumed to 0.9, and battery capacity $B_{max} = 100$ kWh. Coefficients of the battery degradation curve are $\gamma_0 = 6000$, $\gamma_1 = 2.5$, $\gamma_2 = 0.5$ based on [39]. We assume that at each time-slot half of the users are able to provide reserve, and the maximum reserve capacity of each user is uniformly distributed in [0, 40%] of its load. All the simulations are performed on a PC with 2.5 GHz CPU, and the memory is 16 GB.

A. Computational Properties of FCA Mechanism

In this subsection, we illustrate the computational property of the proposed FCA mechanism, in terms of the optimality and computation time. The proposed FCA mechanism includes an approximate algorithm (AA) for the WDP and critical payment (CP) calculation scheme. In contrast, the VCG mechanism includes an exact algorithm for the WDP and the VCG payment scheme.

TABLE I
AVERAGE SOCIAL COST ACHIEVED BY FIVE DIFFERENT ALGORITHMS

No. of Users	Optimal	AA	LR	GA	$\sqrt{M}$
720	90.14	93.53	91.37	181.48	109.41
780	83.36	85.29	84.57	166.40	101.07
840	77.22	79.49	78.13	162.88	93.32
900	69.29	71.43	70.17	149.47	84.05
960	66.35	68.30	67.22	142.87	78.85
1020	64.82	66.16	65.86	141.85	76.55
1080	62.87	64.95	63.74	140.84	75.91

The bold values are the average social cost achieved by our proposed algorithm.

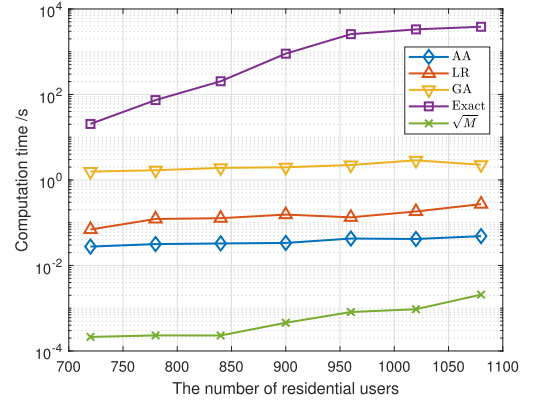


Fig. 4. Computation time comparison of five different algorithms.

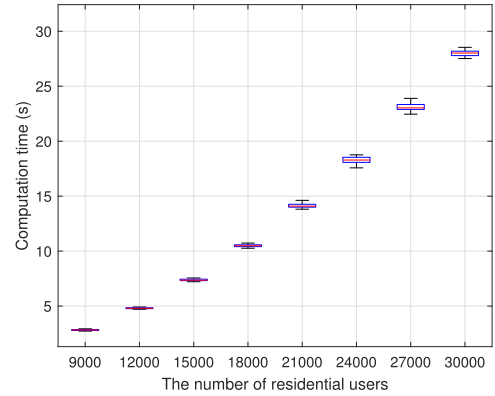


Fig. 5. Computation time of AA with large-scale residential users.

We first compare the objective value achieved by AA and the optimal value of WDP, which is the social cost of reserve procurement. Seven different cases are studied, where the residential user number is increased from 720 to 1080. For each case of use number, 100 random simulations are performed. In each random experiment, the approximation ratio of AA is recorded as a data sample. The approximation ratio is defined as the objective value of WDP obtained by AA divided by the corresponding optimal value of WDP. If the approximation ratio equals to 1, it means the solution is optimal. Note that no algorithm can achieve an approximation ratio of less than 1. Seven box plots in Fig. 3 display the statistical distribution of the recorded approximation ratio data. For each box plot, there are five horizontal lines, where the highest line represents the maximum approximation ratio and the lowest line is the minimum one. The second highest horizontal line and the second lowest

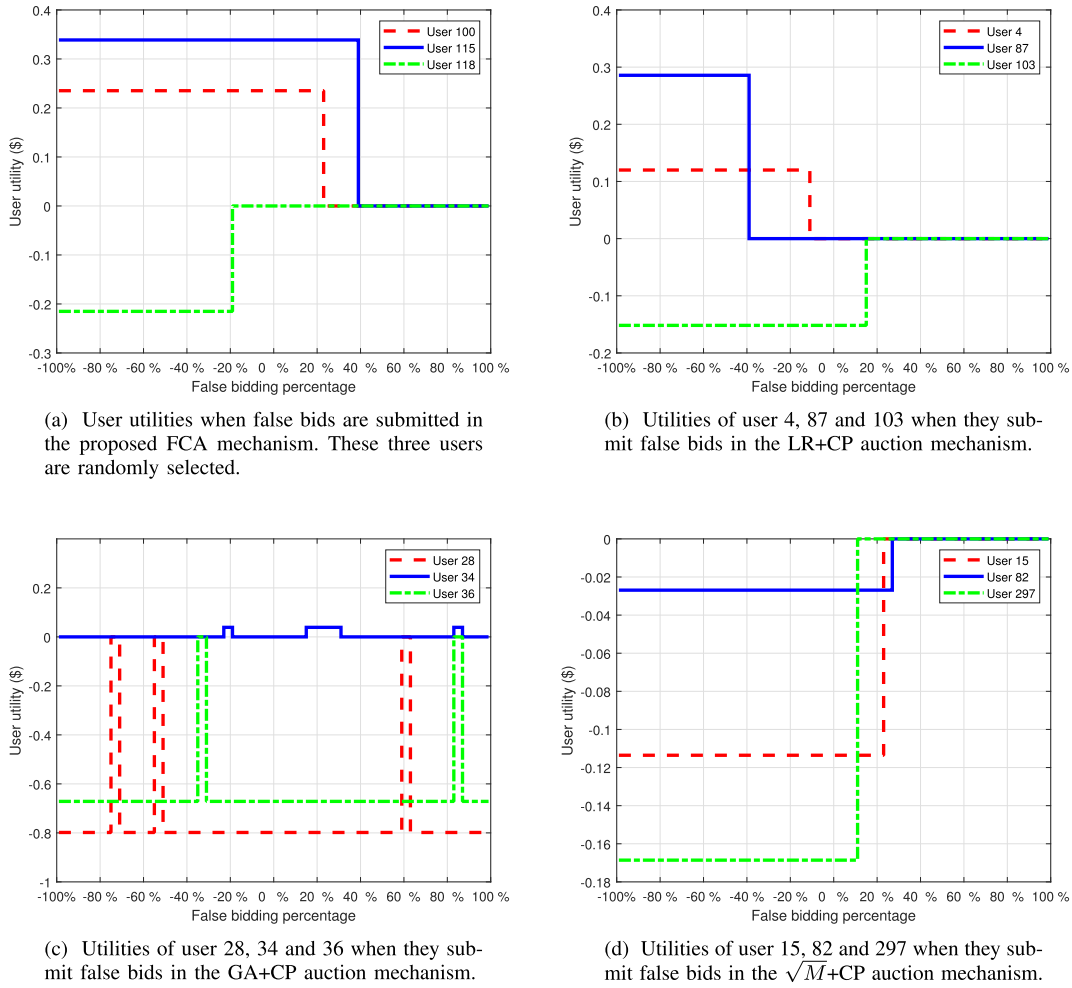


Fig. 6. Truthfulness of different auction mechanisms using the critical payment scheme. In each mechanism, three representative users are selected. For each user, the submitted bid cost is varied, and the resultant utility is plotted. The “false bidding percentage” in the x-axis measures how much the bid cost is changed. 0% means the user tells the truth, 20% means she reports a cost that is 20% higher than the true cost, -20% means she reports a cost that is 20% lower than the true cost.

one constitutes a box together, which means the approximation ratio from the 25th percentile to the 75th percentile of the whole data set. The middle line inside the box is the median of the data. We can conclude that the optimality loss of AA is in general less than 6%. In fact, the average optimality loss is 2.92%.

We further compare the average social cost obtained by the proposed AA and three other kinds of approximation algorithms. They are the genetic algorithm (GA) [29], the Lagrangian relaxation (LR) method used in [47] and [48], and the $\sqrt{M}$ -approximation algorithm [49]. Note that the LR method can be implemented in a distributed manner. The average social costs for different cases of user numbers are presented in Table I. It can be observed that both AA and LR can achieve near-optimal social cost, while GA and $\sqrt{M}$ -approximation have substantial optimality gaps. In general, the social cost decreases with the increase of user number, as more low cost users are available for the reserve procurement.

Meanwhile, the average computation times of five different algorithms are illustrated in Fig. 4, i.e., the proposed AA, LR, GA, $\sqrt{M}$ -approximation and the exact algorithm. The exact algorithm is solving the WDP exactly through the commercial

solver CPLEX. Note that the proposed AA is the second fastest one among five algorithms. Compared with the exact algorithm used in the VCG mechanism, the computation speed of AA can be at least 10^4 times faster when the number of users is above 1000. Therefore, the FCA mechanism can achieve near-optimal social cost with much less computation time than VCG. Despite that the $\sqrt{M}$ -approximation algorithm is faster than AA, its optimality gap is much larger than that of AA.

After that, we further expand the scale of user numbers. Since the computation time of VCG is already more than 1 hour when the number is over 1000, we do not perform VCG in this large-scale case. As shown in Fig. 5, the number of residential users is increased up to 30000, the total computation of AA is still less than 30s. Hence, our proposed FCA mechanism is suitable for application in the large-scale case.

B. Economic Properties of FCA Mechanism

From Fig. 1 and Fig. 4, we can find that other approximate algorithms such as LR can also achieve similar approximation ratio with low computation time. In this subsection, we will

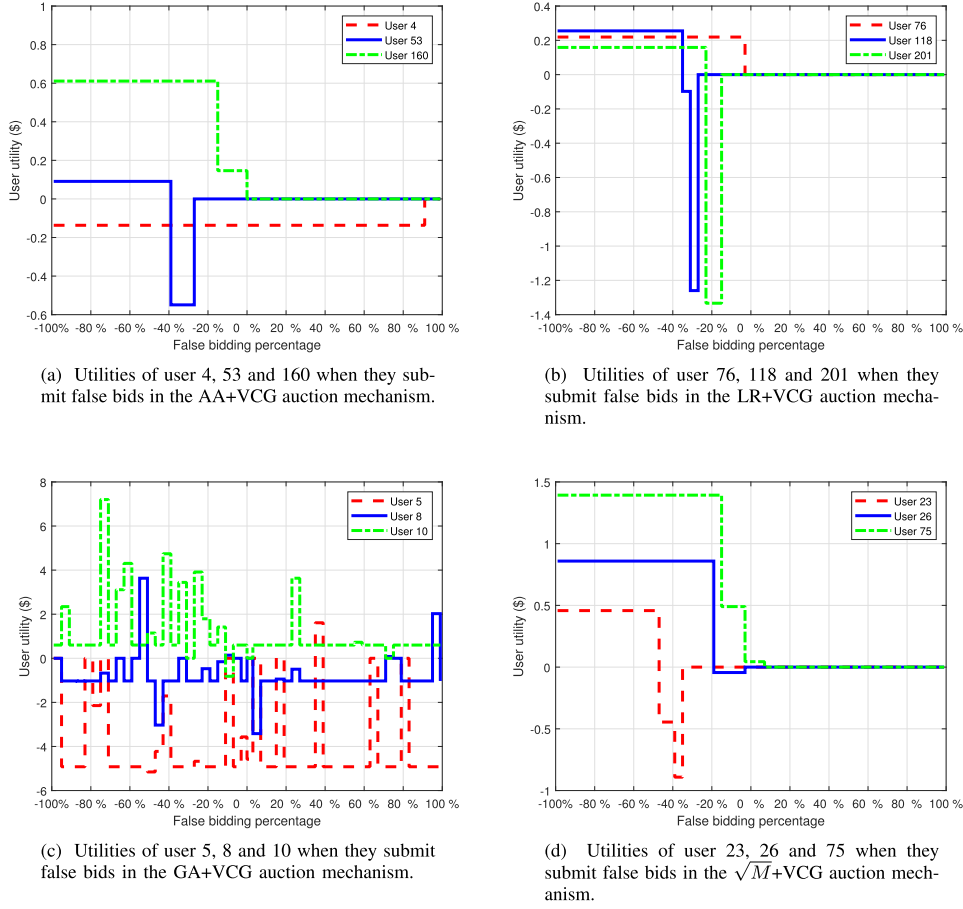


Fig. 7. Truthfulness of different auction mechanisms using the VCG payment scheme. In each mechanism, three representative users are selected. For each user, the submitted bid cost is varied, and the resultant utility is plotted.

illustrate the advantage of FCA over other mechanisms in terms of its economic properties.

Noticing that auction mechanisms have two components: i.e. a WDP and a payment scheme, we compare the truthfulness property of eight different auction mechanisms, which are the combinations of four different WDP algorithms and two payment schemes. The four WDP algorithms are AA, LR, GA and $\sqrt{M}$ -approximation, while the two payment schemes are the critical payment (CP) and VCG scheme. Therefore, as shown in Figs. 6 and 7, the eight mechanisms studied here are AA+CP, LR+CP, GA+CP, $\sqrt{M}$ +CP, AA+VCG, LR+VCG, GA+VCG and $\sqrt{M}$ +VCG, where AA+CP is our proposed FCA mechanism.

In the FCA mechanism case, i.e., Fig. 6(a), three users are randomly selected.³ The utilities of residential users when they report false cost value are shown. Here, 20% in the X-axis means that user k reports a cost $b_k = 120\% * \bar{b}_k$, where $\bar{b}_k$ is her true cost, similarly, -40% means she reports $b_k = 60\% * \bar{b}_k$, and 0% represents that she tells the truth. We can observe that for each user, reporting false value cannot increase her utility, thus validating the truthfulness of FCA.

³Selecting other users can achieve similar results.

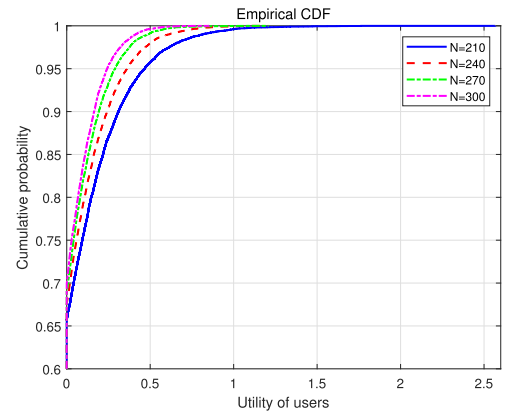


Fig. 8. Cumulative distribution function of each user's utility.

As for other mechanisms, i.e., Figs. 6(b)–6(d) and 7(a)–7(d), for each user, there is always a false cost value that brings larger utility to her than telling the truth. Therefore, the truthfulness property no longer holds. Some users may strategically report false information to the aggregator, compromising the mechanism's efficiency, robustness and fairness [50].

In Fig. 8, the utility distributions of users are presented. It is easy to observe that the utilities of all the users are non-negative,

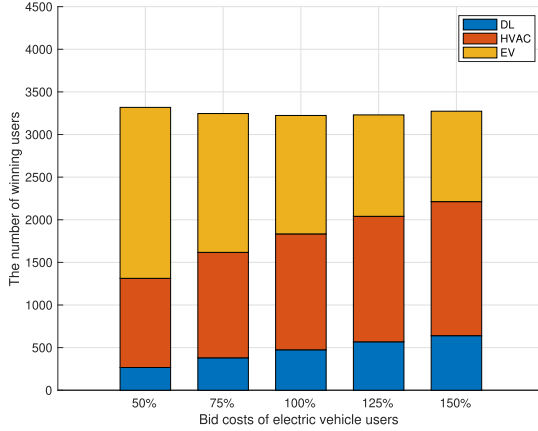


Fig. 9. The number of winning users with three kinds of flexible loads when bid costs of EV users change.

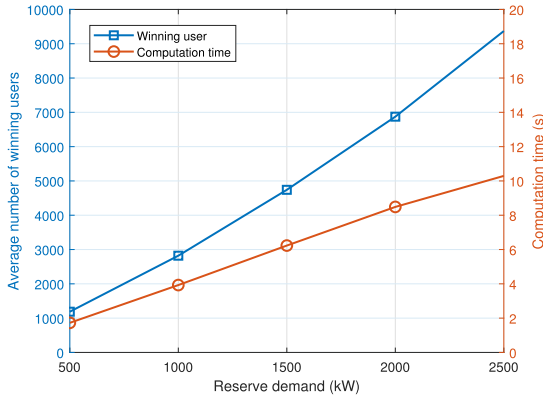


Fig. 10. Average number of winning users and the computation time when the reserve demand of aggregator varies.

verifying the individual rationality of FCA. Taking “N=210” for instance, about 65% of users have zero utility, they are in general losers in the auction, where zero payment and zero cost lead to zero utility. Comparing the distributions of four different cases, we can find that when the number of users increases, the competition between users grows more intense, then the average utilities of winning bids become lower.

C. Sensitivity Results

Finally, some sensitivity experiments are conducted by varying the parameters of the auction. In Fig. 9, we vary the bid cost of all the EV users as {50%, 75%, 100%, 125%, 150%} of their original cost. As the cost of EV users increases, the number of winning EV users decreases, while the population of winning users in the other two categories become larger.

In Fig. 10, when the reserve demand of aggregator increases, more reserve needs to be procured from the residential users. Therefore, the number of winning users also increases. As for the computation time, the number of “while” iterations in Algorithm 1 is the same with the number of winning bids. Thus, the corresponding computation time rises as well.

VII. CONCLUSIONS AND FUTURE WORK

In this paper, we have proposed a novel combinatorial reverse auction framework as the market mechanism for aggregating large-scale residential users in regulation reserve provision. In this auction, the aggregator is the auctioneer and procures reserves from residential users. As for the users, due to the temporal coupling and the coupling between up-reserve and down-reserve, they can submit package bids consisting of combinations of both up and down reserves over several time-slots to fully express their costs. In order to tackle with the computational difficulty brought by the large number of users, an approximate algorithm was developed to determine the winners of the auction. After that, we designed a critical payment scheme to calculate the payment to each winning user. The proposed auction mechanism was rigorously proved to be truthful, individually rational and have polynomial-time computation complexity. Case studies based on deferrable loads, HVACs and EVs were conducted to validate the proposed auction mechanism. Extensive simulations demonstrated the advantages of our proposed mechanism over other related mechanisms. In particular, compared with the widely accepted truthful VCG mechanism, the proposed mechanism is 10^4 times fast than VCG when the user number is above 1000. Meanwhile, it is able to achieve near-optimal social cost, where the average optimality loss is 2.92%.

In the future, we will extend our combinatorial auction mechanism to the “multi-minded” case, where each user are allowed to submit multiple bids to the aggregator. In this way, the users will have more flexibility to participate in the reserve provision services.

APPENDIX A PROOF OF LEMMA 1

We prove this lemma by contradiction. Suppose that bidding $\beta_k = (Q_k, b_k)$ wins in the q -th iteration of Algorithm 1, while bidding with $\hat{\beta}_k = (Q_k, \hat{b}_k)$, $\hat{b}_k \leq b_k$ loses the auction. Since β_k wins in the q -th iteration, according to Line 7 of Algorithm 1, its rank value r_k^q must be the smallest of all the bids remaining in the q -th iteration. By the assumption of $\hat{\beta}_k$ losing, the rank value of $\hat{\beta}_k$ must be higher than that of β_k , i.e., $r(\hat{\beta}_k) > r(\beta_k)$. Note that the bidding sets Q_k of $\hat{\beta}_k$ and β_k are the same, then according to equation (22), $\hat{b}_k > b_k$ holds, which contradicts the assumption that $\hat{b}_k \leq b_k$.

APPENDIX B PROOF OF LEMMA 2

Since Algorithm 1 is monotone by Lemma 1, a value $c_k \geq 0$ exists such that when the bid value of user k lies in the set $[0, c_k]$, she will always win the auction. Define that

$$c(\beta_k) = \max\{c_k \mid \text{bidding } b_k \leq c_k \text{ wins the auction}\}. \quad (25)$$

Assume another bid with value $\hat{c}_k > c(\beta_k)$ is submitted by user k , and she also wins with this bid. From the definition of monotonicity, all bids which have values smaller than $\hat{c}_k$ still win. This violates the assumption that $c(\beta_k)$ is the maximum of c_k , where c_k satisfies that bidding less than c_k always wins the

auction, i.e., (25). Therefore, by this contradiction, the bid with value $\hat{c}_k > c(\beta_k)$ loses.

APPENDIX C PROOF OF LEMMA 3

In Algorithm 2, suppose that in the q_c -th iteration, the critical bid $\beta_k^c = (Q_c, b_k^c)$ wins, denote the unserved up and down reserve as $\bar{L}_{\tau,k}^u$ and $\bar{L}_{\tau,k}^d$ respectively. According to equation (23), the critical payment is

$$\begin{aligned} p_k^c &= r(\beta_k^c) \cdot \sum_{\tau: (q_{\tau,k}^u, q_{\tau,k}^d) \in Q_k} (\min\{\bar{L}_{\tau,k}^u, q_{\tau,k}^u\} + \\ &\quad \min\{\bar{L}_{\tau,k}^d, q_{\tau,k}^d\}) , \\ &= b_k^c \cdot \left\{ \sum_{\tau: (q_{\tau,k}^u, q_{\tau,k}^d) \in Q_k} (\min\{\bar{L}_{\tau,k}^u, q_{\tau,k}^u\} + \right. \\ &\quad \left. \min\{\bar{L}_{\tau,k}^d, q_{\tau,k}^d\}) \right\} / \\ &\quad \left\{ \sum_{\tau: (q_{\tau,k}^u, q_{\tau,k}^d) \in Q_c} (\min\{\bar{L}_{\tau,k}^u, q_{\tau,k}^u\} + \right. \\ &\quad \left. \min\{\bar{L}_{\tau,k}^d, q_{\tau,k}^d\}) \right\} , \end{aligned} \quad (26)$$

where $r(\beta_k^c)$ is the rank value of critical bid β_k^c . Two cases are discussed:

For the first case, suppose the bid of user k is $\hat{b}_k \leq p_k^c$, then its ranking value $r(Q_k, \hat{b}_k)$ in the q_c -th iteration will be

$$\begin{aligned} r(Q_k, \hat{b}_k) &= \hat{b}_k / \sum_{\tau: (q_{\tau,k}^u, q_{\tau,k}^d) \in Q_k} (\min\{\bar{L}_{\tau,k}^u, q_{\tau,k}^u\} \\ &\quad + \min\{\bar{L}_{\tau,k}^d, q_{\tau,k}^d\}) \\ &\leq p_k^c / \sum_{\tau: (q_{\tau,k}^u, q_{\tau,k}^d) \in Q_k} (\min\{\bar{L}_{\tau,k}^u, q_{\tau,k}^u\} \\ &\quad + \min\{\bar{L}_{\tau,k}^d, q_{\tau,k}^d\}) \\ &= r(\beta_k^c), \end{aligned} \quad (27)$$

where the last equality follows from (26) and the definition of rank value r in (22). Thus, bid $(Q_k, \hat{b}_k)$ has higher ranking priority than the critical bid β_k^c . Since β_k^c itself wins in the q_c -th iteration when bid (Q_k, b_k) is absent, $(Q_k, \hat{b}_k)$ will be the winner of the auction.

For the second case, suppose the bid of user k is $\tilde{b}_k > p_k^c$. Now we compare the rank value of $(Q_k, \tilde{b}_k)$ with the rank value of $\beta_{k,q}^h$, which is the head of the nondecreasing list generated in the q -th iteration. In other words, $\beta_{k,q}^h$ is the q -th winning bid in Algorithm 2. Denote the unserved up-reserve demand in the q -th iteration as $\tilde{L}_{\tau,k}^u$, and the unserved down-reserve as $\tilde{L}_{\tau,k}^d$. Since p_k^c is the largest value among all the $p_{k,q}^h$, according to equation (22), we have

$$\tilde{b}_k / \sum_{\tau: (q_{\tau,k}^u, q_{\tau,k}^d) \in Q_k} (\min\{\tilde{L}_{\tau,k}^u, q_{\tau,k}^u\} +$$

$$\begin{aligned} &\min\{\tilde{L}_{\tau,k}^d, q_{\tau,k}^d\}) \\ &> p_{k,q}^h / \sum_{\tau: (q_{\tau,k}^u, q_{\tau,k}^d) \in Q_k} (\min\{\tilde{L}_{\tau,k}^u, q_{\tau,k}^u\} + \\ &\quad \min\{\tilde{L}_{\tau,k}^d, q_{\tau,k}^d\}) \\ &= r(\beta_{k,q}^h), \end{aligned} \quad (28)$$

where the inequality follows from $\tilde{b}_k > p_k^c \geq p_{k,q}^h$, and the equality follows from equation (23). Hence, the rank priority of $(Q_k, \tilde{b}_k)$ is lower than all the winning bids obtained by Algorithm 2, which means that bidding great than $p^c(\beta_k)$ will lose the auction.

Lemma 3 follows from the definition of critical value.

APPENDIX D PROOF OF THEOREM 2

When user k bids truthfully with $\bar{\beta}_k = (Q_k, \bar{b}_k)$, we denote her utility as $\bar{u}_k$. When she submits a bid $\beta_k = (Q_k, b_k)$ with a false value $b_k \neq \bar{b}_k$, her utility is set as u_k . Four different cases are discussed as follows:

- *Bid $\bar{\beta}_k$ wins, while β_k loses:* $\bar{u}_k \geq 0$, because the utility of truthful bidder is non-negative according to Lemma 4. Meanwhile, the utility of losing bid is $u_k = 0$. This yields $\bar{u}_k \geq u_k$;
- *Bid $\bar{\beta}_k$ loses, while β_k wins:* this scenario can only appear when $\bar{b}_k > p_k^c \geq b_k$ by virtue of Lemma 3. As $\bar{\beta}_k$ loses, $\bar{u}_k = 0$. Despite that user k may submit a false bid value b_k and win the auction, the true cost is still $\bar{b}_k$ and her payment is p_k^c which is irrelevant with her bid cost b_k , then we can obtain $u_k = p_k^c - \bar{b}_k < 0$. Clearly, $\bar{u}_k > u_k$;
- *Both bid $\bar{\beta}_k$ and β_k win:* From the payment scheme proposed in Algorithm 2, bid $\bar{\beta}_k$ and β_k receive the same payments, and they also have the same true costs, then $\bar{u}_k = u_k$.
- *Both bid $\bar{\beta}_k$ and β_k lose:* The utility is zero as the received payment and cost are both zero, i.e., $\bar{u}_k = u_k = 0$.

To summarize, bidding truthfully is always a dominant strategy with $\bar{u}_k \geq u_k$ holds for all the different cases.

APPENDIX E PROOF OF THEOREM 3

First we consider Algorithm 1. The “for” loops in Step 3 have time complexity $O(K)$, while the one in Step 9 is $O(T)$. The complexity of the sorting operation in Step 6 is $O(K \log K)$. Since each iteration of the “while” loop in Step 2 removes one bid, then the “while” loop will continue for at most K times. Considering K is usually much larger than T , the total complexity is $O(K^2 \log K)$. As for Algorithm 2, the results are similar. Therefore, we have proved this theorem.

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